Entropy Jumps and Information Relaxation in Financial Markets: from maximum entropy in volume time to entropy-calibrated stress testing

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Abstract

This paper rebuilds the information-theoretic account of price formation around a single asymmetry. We first close the bridge from maximum entropy to Gaussian prices where it is usually left open, by defining the price as accumulated order flow and applying the maximum-entropy principle in *volume time* rather than calendar time: with the traded volume fixing the variance budget, the least-committal conditional distribution of the log-price increment is Gaussian, and temporal consistency then makes the log-price a Brownian motion subordinated to the volume clock, hence the price a geometric Brownian motion. Read backwards, the same argument says that non-Gaussian prices are the signature of a conditional distribution that is *not* entropy-maximal – that is, of binding information.

We then decompose the entropy of a return vector into three channels: scale ($\sum_i \log \sigma_i$), dependence ($\frac{1}{2}\log\det R \leq 0$) and shape (the entropy deficit $\mathcal{D} = D_{KL}(p \,\|\, q)$, which is skewness and fat tails). Combining the last two gives a scale-free structural index $\mathcal{J}_t = \mathcal{D}_t - \frac{1}{2}\log\det R_t = D_{KL}(p_t \,\|\, \prod_i q_{i,t}) \geq 0$, which measures how far the market is from the maximum-entropy reference at unchanged volatility. Because conditioning cannot raise expected entropy, an information event lowers the entropy ceiling by exactly the mutual information of the news: downward jumps in entropy are a theorem, not an assumption. What we postulate is the other half – that entropy relaxes back diffusively (P1), and that the constraint news imposes is one-sided, so a jump carries negative skewness (P2). Together these give $\mathcal{J}$ a jump-diffusion of Barndorff-Nielsen–Shephard type: up in jumps, down by diffusion.

Seven hypotheses are tested on 33 years of daily individual-stock returns (twenty S&P 500 constituents, 1990–2022), replicated on the `EuStockMarkets` panel, and — this is the part that matters — judged against two calibrated nulls built by resampling the observed returns: one i.i.d., which keeps the fat marginal tails and destroys all temporal structure, and one in 21-day blocks, which additionally keeps volatility clustering. Taken at face value the results look strong: stress raises $\mathcal{J}$ by 2.89 nats, changes in $\mathcal{J}$ are overwhelmingly one-sided (158 up-jumps against 9 down, skewness +12.3), up-jumps carry negative market skewness, and the dependence and tail channels lock together under stress (0.04 calm against 0.66). Against the nulls, most of that evaporates. Jump asymmetry, the skewness signature and the channel coupling are all reproduced — several of them *exceeded* — by resampled data, because a non-negative convex functional of estimated moments rises sharply and decays smoothly whenever a fat-tailed observation enters the estimator. Exactly two statistics clear both nulls, and they are the ones worth keeping: the regime difference in the *dependence* channel (−1.72 nats against −0.29 and −0.64), and the range of $\mathcal{J}$ over the sample (18.95 against 10.49 and 14.23). What distinguishes a crisis from a run of large returns is that the cross-section couples — the effective number of independent risk modes falls from 13.2 to 10.5 out of 20 — not that the marginals fatten. The relaxation half-life is not identified: at 120 days it lies above the i.i.d. null (92) and below the block null (162). Two further results run against the paper's interest: the volatility–dependence

compensation reported for industry portfolios does not replicate on individual firms, and no entropy channel improves out-of-sample forecasts of forward risk over a volatility benchmark.

The applied contribution is a stress-testing construction. A scenario of severity η is the worst conditional distribution within a Kullback–Leibler ball of radius η nats around the market's current conditional distribution; the radius is calibrated on the market's own realised information flow $D_{KL}(p_t \,\|\, p_{t-h})$, so severity carries a return period rather than an adjective, and the mean shift and variance inflation are drawn on one budget rather than shocked piecemeal. One nat buys $\sqrt{2}$ sigma. The same metric prices existing scenarios: a textbook bundle – volatilities doubled, all correlations forced to 0.9, a three-sigma shock – costs 52 nats against 13 nats for the one-in-fifty-year rung, and we decompose that price to show where it goes.

Contents

1 Introduction

Two facts about financial prices sit awkwardly together. The first is that the Gaussian model is the natural answer to a question about ignorance: fix a mean and a variance, know nothing else, and the maximum-entropy distribution is Gaussian. The second is that returns are not Gaussian – they are fat-tailed, negatively skewed, and their dependence tightens exactly when it matters.

The usual reaction is to treat the second fact as a defect of the first model and to patch it: add jumps to the price, add a stochastic volatility factor, add a copula. This paper takes the opposite route. If the Gaussian is what maximum entropy delivers, then every departure from it is a measurement of how far the market is from maximal ignorance – that is, a measurement of information. The question is then not how to patch the Gaussian but how to read the size, dynamics and direction of the gap.

The paper is built around one asymmetry in that gap. Information arrives in lumps and is absorbed gradually. A central-bank surprise, a default, a war removes uncertainty in an instant; the market then spends weeks working out what it implies, and the uncertainty rebuilds. If entropy is the right currency for uncertainty, its dynamics should be sawtoothed: sharp falls, slow recoveries. That shape is not a stylised fact anyone has had to assume – half of it is a theorem. Conditioning cannot raise expected entropy, so the arrival of a signal lowers the entropy ceiling by exactly the mutual information between the signal and the price. The size of the jump, in nats, *is* the information content of the news. What has to be postulated is the recovery, and the direction.

Three contributions follow.

The volume bridge (Section 2). The step from "maximum entropy gives a Gaussian" to "prices are Gaussian" is routinely asserted and rarely argued, because maximum entropy is a statement about a distribution with a fixed variance and says nothing about where the variance comes from. We close it by defining the price as accumulated order flow and applying the principle in volume time: total traded volume, which is observable, fixes the variance budget, and the maximum-entropy increment conditional on volume is Gaussian with variance proportional to volume. This is the mixture-of-distributions hypothesis (Clark, 1973; Tauchen & Pitts, 1983; Ané & Geman, 2000) derived rather than assumed. Temporal consistency then makes the log-price a Brownian motion subordinated to the volume clock, and Itô's lemma makes the price a geometric Brownian motion. The bridge also explains, in the same sentence, why calendar-time returns are not Gaussian: either the conditional distribution is not entropy-maximal (information binds), or the volume clock is random, or both – and the two sources leave different fingerprints.

Three channels and a scale-free index (Section 3). The Gaussian entropy of a return vector splits exactly into a scale term and a dependence term, and the true entropy differs from it by the entropy deficit $\mathcal{D} = D_{KL}(p \,\|\, q)$, which is where skewness and fat tails live. The scale term dominates the level of differential entropy and is uninformative about information; stripping it out leaves $\mathcal{J} = \mathcal{D} - \frac{1}{2}\log\det R$, which is exactly $D_{KL}(p \,\|\, \prod_i q_i)$ – the divergence from a product of Gaussian marginals with the same scales. $\mathcal{J}$ is non-negative, invariant to the level of volatility, and additive across its two entropy-destroying channels. It is the state variable the rest of the paper is about. Empirically it shares only 21% of its variance with volatility, and its shape channel is essentially orthogonal to volatility (corr = 0.09).

Stress testing in nats (Section 8). A scenario of severity η is the worst conditional distribution within a KL ball of radius η around today's. The construction is not new (Glasserman & Xu, 2014; Breuer & Csiszár, 2013); the calibration is. The radius is read off the market's own realised information flow, the h-step revision $D_{KL}(p_t \,\|\, p_{t-h})$, which is measured in the same unit as the radius, so a severity acquires a return period. Because the mean shift and the variance inflation are chosen on one budget, the scenario cannot be internally inconsistent, and because *any* scenario can be priced on the same scale, "is this stress test severe enough?" becomes arithmetic.

A warning about measuring any of this. The structural index is a non-negative, convex functional of estimated second, third and fourth moments. Estimation noise on fat-tailed data pushes such a functional *up* sharply and lets it decay *smoothly*, which is precisely the sawtooth the postulates predict. Every jump and relaxation statistic below is therefore reported against a null built by i.i.d. resampling the observed returns — keeping the marginal fat tails, destroying the timing. The result is uncomfortable and, we think, the most useful thing in the paper: the jump asymmetry, which looks overwhelming, does not survive; the persistence and amplitude do. Anyone building a crisis indicator from a convex functional of estimated moments should run this null before believing their own figure.

Section 4 states the theorem and the two postulates and derives the jump-diffusion; Section 5 lists the hypotheses; Sections 6–7 give data, estimators and results, including a placebo null that the jump and relaxation statistics must clear; Section 8 builds the stress test; Section 9 is honest about what none of this establishes.

2 From order flow to a Gaussian price: the volume bridge

2.1 The price as accumulated order flow

Over $[t, t+\Delta]$ let the market execute M trades, trade k carrying signed volume $\epsilon_k v_k$ with $\epsilon_k \in \{-1, +1\}$ the sign (buyer- or seller-initiated) and $v_k > 0$ the size. Write $V = \sum_k v_k$ for total volume and $Q = \sum_k \epsilon_k v_k$ for net signed order flow.

Assumption 1 (Linear impact). *The log-price responds linearly to net signed order flow, $\Delta X_{t,\Delta} = X_{t+\Delta} - X_t = \lambda Q$, with $\lambda > 0$ constant over the interval.*

This is Kyle's (1985) equilibrium: the market maker, unable to distinguish informed from uninformed flow, sets a linear pricing rule and λ is the inverse depth. It is the definition of "price as a function of traded volume" that the rest of the section needs, and it is the weakest link in the chain; Section 2.5 says what happens when it fails.

Assumption 2 (Exchangeable signs and a volume-proportional variance)**.** *Conditional on V and on the information available at t, the market's belief about Q is constrained only by its first two moments, with $\mathbb{E}[Q \mid V] = m$ and $\mathrm{Var}(Q \mid V) = \varsigma^2 V$.*

The variance form is what one gets when trade signs are exchangeable with bounded correlation and sizes have a stable mean: for independent signs, $\mathrm{Var}(Q) = \sum_k v_k^2 \approx \bar{v}V$. The economic content is that *volume, not time, is the natural measure of how much can happen.*

2.2 Maximum entropy in volume time

Proposition 1 (Maximum entropy under first and second moments)**.** *Among absolutely continuous distributions on $\mathbb{R}^N$ with mean μ and covariance $\Sigma \succ 0$, the Gaussian $\mathcal{N}(\mu, \Sigma)$ uniquely maximises differential entropy, and $h^{\max}(\Sigma) = \frac{1}{2}\log\big((2\pi e)^N \det\Sigma\big)$.*

Proof. Let $q = \mathcal{N}(\mu, \Sigma)$ and let p have the same first two moments and finite differential entropy. Since $\log q$ is an exact quadratic, its expectation depends only on those moments, so $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q] = -h(q)$. Non-negativity of relative entropy gives $0 \le D_{KL}(p \,\|\, q) = -h(p) + h(q)$, hence $h(p) \le h(q)$, with equality iff $p = q$ almost everywhere. The argument requires no differentiability of p and no prior existence of a maximiser. □

The variational route – a Lagrangian in the normalisation, mean and covariance constraints, whose first-order condition makes $\log p$ quadratic – explains *why* the maximiser is Gaussian, but the divergence argument is what establishes optimality and uniqueness.

Applying Proposition 1 to Assumption 2 in one dimension and mapping through Assumption 1:

$$\Delta X_{t,\Delta} \;\big|\; V \;\sim\; \mathcal{N}\big(\lambda m,\ \lambda^2\varsigma^2 V\big)\,. \tag{1}$$

Equation (1) is the bridge. Three things about it deserve emphasis.

First, *the conditioning variable is volume, not time.* Maximum entropy does not deliver a variance; it delivers the least-committal distribution given one. Assumption 2 supplies the variance from an observable, and that observable is traded volume. Equation (1) is the mixture-of-distributions hypothesis of Clark (1973) and Tauchen & Pitts (1983), obtained here as a maximum-entropy statement rather than posited; Ané & Geman (2000) show empirically that returns measured in trade time are close to Gaussian even though calendar-time returns are not, which is the prediction (1) makes.

Second, *Gaussianity is conditional.* Unconditionally, ΔX is a scale mixture of normals over the random volume V, hence leptokurtic even when (1) holds exactly. This is the Mandelbrot & Taylor (1967) subordination mechanism, and it means fat tails are not by themselves evidence against maximum entropy.

Third, the multivariate extension is immediate: with N assets, $\Delta\mathbf{X} \mid V \sim \mathcal{N}(\boldsymbol{\mu}V, \Sigma V)$, and Proposition 1 applies as stated.

2.3 From Gaussian increments to Brownian motion and to GBM

Let $\theta_t = \int_0^t v_s\,ds$ be the cumulative volume clock. Equation (1) gives the one-increment distribution; a process needs more.

Assumption 3 (Temporal consistency)**.** *In the volume clock, increments over disjoint intervals are independent, sample paths are continuous, and the covariance of an increment is proportional to elapsed volume.*

Under Assumption 3 with constant coefficients there is a matrix B with $BB^\top = \Sigma$ such that $X_\theta = X_0 + \mu\theta + BW_\theta$ in volume time, so in calendar time

$$X_t = X_0 + \mu\,\theta_t + B\,W_{\theta_t}, \tag{2}$$

a Brownian motion *subordinated* to the volume clock. If trading intensity is constant, $\theta_t = \bar{v}\,t$ and (2) is ordinary Brownian motion in calendar time; exponentiating and applying Itô,

$$\frac{dS_{i,t}}{S_{i,t}} = \big(a_{i,t} + \tfrac{1}{2}\sigma_{i,t}^2\big)\,dt + \sigma_{i,t}\,dW_{i,t}, \tag{3}$$

so the drift of the price is not the drift of the log-price – the same second-order correction that will reappear in Section 3 when we differentiate $\log\det R_t$.

It is worth being precise about what Assumption 3 does. By Monroe's (1978) theorem, *every* continuous local martingale is a time-changed Brownian motion, so subordination alone has no empirical content. The content is in the clock: (1) says the right clock is volume, and Assumption 3 says the market has no memory in that clock. Maximum entropy is what selects Brownian motion from the class Monroe's theorem allows.

2.4 The contrapositive

Proposition 1 runs both ways. Write q_t for the Gaussian with the conditional mean and covariance of p_t. Then

$$\mathcal{D}_t \;\equiv\; h(q_t) - h(p_t) \;=\; D_{KL}(p_t \,\|\, q_t) \;\geq\; 0, \tag{4}$$

with equality if and only if p_t is Gaussian. So: *prices are non-Gaussian exactly to the extent that their conditional distribution is not entropy-maximal*, and $\mathcal{D}_t$ measures the extent. In the language of Section 2, a positive $\mathcal{D}_t$ means constraints beyond the first two moments are binding – that somebody knows something the second moment does not encode.

Two sources produce a positive deficit, and separating them matters. A random volume clock produces a scale mixture, which is symmetric and leptokurtic. Binding directional information produces asymmetry. Section 3 splits $\mathcal{D}$ along exactly this line.

2.5 Where the bridge is weak

Assumption 1 is linear; the empirical impact of a metaorder is closer to square-root in size, which makes the mapping from flow to price concave and the induced return distribution thinner-tailed than the flow. Assumption 2 treats signs as exchangeable; order flow is in fact strongly autocorrelated, which inflates $\mathrm{Var}(Q)$ above $\varsigma^2 V$ and makes the volume clock an imperfect variance proxy. Neither failure changes the *form* of the argument – maximum entropy given a variance budget is still Gaussian – but both mean the volume clock is an approximation to the information clock rather than the thing itself. We use volume-time reasoning to motivate the framework and calendar-time estimation throughout the empirical work, so no result below depends on (1) holding exactly.

3 Three channels of entropy

3.1 Scale and dependence

Write $\Sigma_t = D_t R_t D_t$ with $D_t = \mathrm{diag}(\sigma_{1,t}, \ldots, \sigma_{N,t})$. Then $\det \Sigma_t = \left(\prod_i \sigma_{i,t}^2\right) \det R_t$ and

$$h(q_t) \;=\; \underbrace{\tfrac{N}{2}\log(2\pi e)}_{\text{constant}} \;+\; \underbrace{\sum_i \log \sigma_{i,t}}_{H_t^{vol}} \;+\; \underbrace{\tfrac{1}{2}\log\det R_t}_{H_t^{dep}}. \tag{5}$$

By Hadamard's inequality $H_t^{dep} \le 0$, with equality iff $R_t = I$: *correlation always destroys entropy*, and $-H_t^{dep}$ is precisely the Gaussian multi-information (total correlation) of the return vector. A useful reading of the same quantity is spectral: with λ_k the eigenvalues of R_t and $\tilde{\lambda}_k = \lambda_k/N$, the participation number $N_t^{\mathrm{eff}} = \exp\!\left(-\sum_k \tilde{\lambda}_k \log \tilde{\lambda}_k\right)$ is the effective number of independent risk modes; it equals N for $R = I$ and falls toward 1 as one collective mode absorbs the variance.

Equation (5) is the ceiling, not the entropy. It is what the covariance matrix alone implies, and Proposition 1 says the truth sits below it.

3.2 Shape: the entropy deficit and what it is made of

Subtracting (4) from (5) gives the paper's decomposition, three channels and a constant:

$$h(p_t) \;=\; \tfrac{N}{2}\log(2\pi e) \;+\; \underbrace{H_t^{vol}}_{\text{scale}} \;+\; \underbrace{H_t^{dep}}_{\text{dependence}} \;-\; \underbrace{\mathcal{D}_t}_{\text{shape}}. \tag{6}$$

The shape channel is skewness and fat tails, and the Edgeworth expansion says so explicitly: for a standardised variable with skewness S and excess kurtosis K near Gaussianity,

$$J \;\approx\; \frac{S^2}{12} + \frac{K^2}{48}. \tag{7}$$

Equation (7) is the answer to the question "where would an entropy jump be visible?": in the third and fourth moments, additively, with an asymmetry channel and a tail channel. It is also the source of the sign problem that Section 4 has to solve, because S enters squared – *negentropy cannot tell a crash from a melt-up*. As an estimator (7) is unusable on daily returns; Section 6.2 documents how badly and what we use instead.

3.3 The structural index

The scale channel is a nuisance for our purposes: it dominates the level of (6) numerically, it moves with volatility, and a change in units changes it. The other two channels are scale-free. Combining them, and using the exact chain rule $D_{KL}(p \,\|\, \prod_i q_i) = \sum_i D_{KL}(p_i \,\|\, q_i) + I(x)$ with I the multi-information:

Definition 1 (Structural information index)**.**

$$\mathcal{J}_t \;\equiv\; \mathcal{D}_t \;-\; H_t^{dep} \;=\; D_{KL}(p_t \,\|\, \textstyle\prod_i q_{i,t}) \;\ge\; 0, \tag{8}$$

the divergence of the conditional joint return distribution from a product of Gaussian marginals with the same scales.

$\mathcal{J}$ is zero if and only if returns are independent and Gaussian; it rises when assets couple, when marginals fatten, or when they skew. It is measured in nats, it is invariant to the level of volatility, and it is additive across its two channels, so one can always ask how much of a move came from dependence and how much from shape. It is the state variable of Section 4.

4 Information dynamics: one theorem, two postulates

4.1 Downward jumps are a theorem

Proposition 2 (Information cannot raise expected entropy)**.** *Let X be the return vector and Y a signal. Then $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y) \leq h(X)$, with equality iff $X \perp Y$. Equivalently, in maximum-entropy terms: enlarging the set of constraints a distribution must satisfy shrinks the feasible set, so the attainable maximum entropy weakly falls.*

Proposition 2 is standard (Cover & Thomas, 2006) and it is the whole of the downward half of the postulated dynamics. It says three useful things. The fall is *discontinuous*, because the signal arrives at a point in time. Its *size in nats is the mutual information of the news* – the jump magnitude is a measurement of how informative the event was, on a scale that does not depend on the asset, the currency or the units. And it is an *average* statement: for a particular realisation y, $h(X \mid Y = y)$ can exceed $h(X)$, because a sufficiently surprising signal can leave the market more uncertain than before. We should therefore expect the jump component to be predominantly, not exclusively, one-sided – which is what Section 7 finds.

4.2 Two postulates

Postulate 1 (Relaxation)**.** Absent new information, the structural index decays toward a long-run level at a constant rate: $d\mathcal{J}_t = -\kappa\left(\mathcal{J}_t - \bar{\mathcal{J}}\right)dt + \eta\, dW_t$ with $\kappa > 0$.

The economics behind P1 is that a constraint goes stale. News is priced in, disagreement reforms, positioning disperses, and the cross-section recovers the freedom the shock removed. Nothing in information theory delivers this; it is the paper's substantive assumption, and κ is the object to be estimated. It is not innocuous: an estimator with memory will manufacture a decay, so Section 6.3 measures the decay a memoryless world would already produce.

Postulate 2 (Asymmetry of the constraint)**.** The constraint information imposes is one-sided in the loss direction, so an entropy jump arrives with negative skewness of the market factor.

P2 is needed because $\mathcal{J}$ is even in skewness and cannot supply a direction. Its content is a statement about liquidity supply: on bad news market-makers and value buyers withdraw on the sell side while the buy side is unaffected, so the same quantity of information produces a larger move down than up. It is separately testable (H5) and separately refutable.

4.3 The resulting dynamics

Proposition 2 with P1 gives a non-Gaussian Ornstein–Uhlenbeck process with one-sided jumps, of the class Barndorff-Nielsen & Shephard (2001) introduced for stochastic volatility:

$$d\mathcal{J}_t \;=\; -\kappa\left(\mathcal{J}_t - \bar{\mathcal{J}}\right)dt \;+\; \eta\, dW_t \;+\; \int_{z>0} z\, N(dt, dz), \tag{9}$$

with N the arrival process of information events and z the mutual information they carry. Read on the entropy itself rather than on $\mathcal{J}$, the sign flips: entropy falls in jumps and diffuses upward.

The observable predictions are sharp, and they are what Section 5 tests: the distribution of $\Delta\mathcal{J}$ is right-skewed; the large moves are up and the small moves are down; and after an up-jump the index decays with a half-life $\ln 2/\kappa$.

4.4 Why fat tails, correlation and jumps arrive together

The framework makes one prediction that neither a volatility model nor a correlation model makes on its own. An information event is, by construction, a *common* constraint: the same news binds every asset in the cross-section. It therefore has to show up in both entropy-destroying channels at once – in dependence, because a common constraint couples the assets, and in shape, because a constraint that binds asymmetrically fattens the loss tail of each marginal. In calm markets, by contrast, the two channels are driven by unrelated idiosyncratic mechanisms and need not comove at all.

The testable form is a *regime-dependent coupling*: the correlation between $-\Delta H^{dep}$ and $\Delta\mathcal{D}$ should be near zero in calm markets and strongly positive in stress. This is H6, and it is where the paper's answer to "does the link between fat tails, correlation and entropy jumps need an extra postulate?" lies: it does not. Given Proposition 2, the coupling follows from the news being common. Only the *sign* needs P2.

5 Hypotheses

H1 (regime signature). In stress regimes the dependence channel falls ($H^{dep}\downarrow$), the shape channel rises ($\mathcal{D}\uparrow$) and therefore the structural index rises ($\mathcal{J}\uparrow$).

H2 (ceiling compensation). ΔH^{vol} and ΔH^{dep} are negatively associated, so the Gaussian ceiling H^{cov} is more stable than its components. This is the hypothesis an earlier draft of this project supported on industry portfolios; it is retained here to be tested on individual firms.

H3 (jump asymmetry). Changes in $\mathcal{J}$ are one-sided: at a common threshold, up-jumps outnumber and outweigh down-jumps, and $\Delta\mathcal{J}$ is right-skewed — *beyond what the same estimator produces on data with the same marginal distribution and no temporal structure.* The qualifier is the whole hypothesis; without it the statement is untestable.

H4 (relaxation). $\mathcal{J}$ mean-reverts with $\kappa > 0$ and a half-life materially longer than the memory of the estimator itself, judged against the same null.

H5 (skewness signature). Up-jumps in $\mathcal{J}$ coincide with negative market skewness and with a rise in the asymmetry channel (postulate P2).

H6 (channel coupling). The dependence and shape channels are weakly coupled in calm markets and strongly coupled under stress.

H7 (incremental value). $\mathcal{J}$, or an indicator built from its jumps, improves out-of-sample forecasts of forward realised volatility, drawdown or worst loss relative to a volatility-only benchmark.

6 Data and estimation

6.1 Datasets

The primary panel is twenty S&P 500 constituents at daily frequency, 1990-01-03 to 2022-12-28 (8,312 return days; 8,061 rolling dates after a 252-day window), obtained from the `sp500` dataset distributed with the `skfolio` package. These are *individual firms*, not industry portfolios. That choice is deliberate: aggregating firms into industries imposes common-factor dependence mechanically, and an earlier draft of this project listed an individual-stock analysis as the robustness test it most needed. The span covers the 1997–98 Asian and LTCM episodes, the dot-com crash, the global financial crisis, the euro-area debt crisis, 2015–16, 2018, COVID-19 and the 2022 selloff.

The replication panel is `EuStockMarkets` (Bollerslev & Ghysels, 1996): daily closing levels of the DAX, SMI, CAC and FTSE, 1,860 contemporaneous trading days from 1991 to 1998, $N = 4$. It is a different market, a different decade and a different cross-sectional size, which is what makes it useful; it is far too small to carry a result on its own.

A reader with access to the Kenneth French 49-industry daily file can reproduce every table on it via `config/default.yaml`; the industry results quoted below for comparison come from the earlier draft of this project.

6.2 Estimators

Covariance. Two estimators run in parallel. The baseline is a 252-day rolling window with Ledoit–Wolf shrinkage (Ledoit & Wolf, 2004), which reproduces the earlier draft's level statistics. The responsive estimator, used for everything involving jumps, is exponentially weighted with a 60-day half-life plus a 10% shrinkage toward the diagonal. The choice matters: with a box-car window, an extreme observation *leaving* the sample produces a discontinuity in the entropy series on a date when nothing happened, which would contaminate any jump test. Exponential weights remove that artefact by construction.

Negentropy. The shape channel is estimated marginally, asset by asset, using Hyvarinen's (1998) bounded-contrast approximation, whose two terms split naturally into an odd (asymmetry) and an even (tail-weight) channel. The Edgeworth form (7) is retained for interpretation only, because it fails badly as an estimator exactly where financial data live. Table 1 makes the failure explicit against a non-parametric reference.

Two limitations are worth stating rather than burying. Estimating the marginal deficits and adding the linear dependence term captures marginal non-Gaussianity and Gaussian-copula dependence but omits non-linear tail dependence; we measure that separately as an average pairwise co-exceedance rate. And Hyvarinen's approximation, while accurate in the leptokurtic region, understates the negentropy of a mildly fat-tailed $t(8)$ by about half, so the levels reported below are conservative.

Jump detection. Jumps in $\mathcal{J}$ are flagged when $|\Delta \mathcal{J}_t|$ exceeds four times a trailing local scale built from bipower variation (Barndorff-Nielsen & Shephard, 2004), which is consistent for the diffusive scale even in the presence of jumps and is lagged by one day so that the threshold judging a date never contains it.

Inference. A 252-day rolling window makes consecutive observations share 251 of 252 underlying return days. Every regime comparison therefore uses a circular block bootstrap with 252-day blocks

Distribution	S	K	J (Vasicek)	J Edgeworth	J Hyvarinen
Gaussian	-0.00	0.03	0.012	0.000	0.000
Uniform	0.00	-1.20	0.179	0.030	0.065
Skew-normal ($a = -4$)	-0.80	0.67	0.074	0.062	0.049
Laplace	-0.01	2.95	0.087	0.181	**0.087**
Student $t(8)$	-0.02	1.63	0.030	0.056	0.016
Student $t(5)$	0.10	5.13	0.061	0.549	**0.052**
Student $t(4)$	-0.00	18.34	0.102	**7.006**	**0.105**
Lognormal ($s = 0.5$)	1.76	5.78	0.203	0.954	0.167
Exponential	2.00	6.11	0.424	1.112	0.268

Table 1: Marginal negentropy in nats, n = 200,000 draws. The reference is the Vasicek spacing estimator of differential entropy. The Edgeworth form $S^2/12 + K^2/48$ overstates a Student-$t(4)$ by a factor of 69; Hyvarinen's approximation is within 3% of the truth there. Daily equity returns sit squarely in the $t(4)$–$t(5)$ region, which is why every deficit reported below uses the latter. Reproduced by `scripts/negentropy_benchmark.py`.

under a null that breaks the value–label link while preserving both the persistence of the series and the clustering of the regime indicator. Predictive comparisons use expanding-window walk-forward estimation refitted annually, evaluation on non-overlapping dates, and a Diebold–Mariano test (Diebold & Mariano, 1995) on the squared-error differential.

6.3 The placebo null

The structural index is a non-negative, convex functional of estimated second, third and fourth moments. Estimation noise alone pushes it *up* sharply and lets it decay *smoothly* – which is exactly the signature (9) predicts. A jump-asymmetry statistic or a relaxation half-life is therefore evidence only relative to that mechanical null, never relative to zero.

We measure the null in two ways. First, a controlled simulation (`scripts/estimator_memory.py`): i.i.d. Gaussian returns with constant covariance – a world with no dynamics whatsoever – run through the production pipeline. A single injected stress day produces a 2.13-nat spike in $\mathcal{J}$ that decays with a half-life of **35 trading days**, purely from the estimator's own memory; on clean data with no event at all, the four-sigma detector flags **37 up-jumps and zero down-jumps** and $\Delta\mathcal{J}$ has skewness +2.9. The mechanical asymmetry is real and it is large.

Second, two calibrated nulls (`scripts/placebo_null.py`) that resample the rows of the observed return matrix, preserving the cross-sectional dependence and the fat marginal tails of real data. The i.i.d. version destroys everything the postulates are about — volatility clustering, time-varying dependence, the arrival dynamics of information. The block version resamples 21-day blocks, so short-run clustering survives and only the longer-horizon regime dynamics are destroyed. Twenty replications of each give the null distributions reported in Table 2. Both apply the identical volatility filter to define stress regimes, so the regime comparisons of H1, H2 and H6 are tested on the same footing as they are estimated.

7 Results

7.1 What the calibrated null leaves standing

Table 2 is the paper's central empirical result and it is uncomfortable. Two nulls are built, both by resampling the observed return matrix and running the identical pipeline — including the identical volatility filter that defines the stress regime — twenty times each.

The i.i.d. null resamples rows independently. It keeps the marginal fat tails and the cross-sectional dependence and destroys *all* temporal structure. It asks: beyond the fact that turbulent windows contain large returns, does the measure detect anything?

The block null resamples circular blocks of 21 days. It additionally keeps short-run volatility clustering and destroys only the longer-horizon regime dynamics. It asks the harder question: beyond clustering, does the measure detect information dynamics?

Neither null is nested in the other in a way that makes one uniformly conservative, and the two disagree often enough that reporting only one would be misleading. A statistic that clears both is evidence; a statistic that clears one and fails the other is not identified.

			i.i.d. null		block-21 null		
	Statistic	Obs.	mean	5–95%	mean	5–95%	Verdict
H1	H^{dep} gap	−1.72	−0.29	−0.46–−0.14	−0.64	−1.14–−0.37	**clears both**
H1	$\mathcal{J}$ gap	2.89	2.32	1.76–2.74	2.21	1.51–3.02	i.i.d. only
H1	$\mathcal{D}$ gap	0.48	1.35	0.98–1.64	0.74	0.30–1.13	fails both
H2	$\mathrm{corr}(\Delta H^{vol}, \Delta H^{dep})$	0.10	0.14	−0.29–0.47	0.16	−0.23–0.57	fails both
H3	up-jumps	158	147.7	139–157	129.5	117–148	block only
H3	down-jumps	9	3.2	2–5	4.2	0–7	*more* than both
H3	mean up-jump (nats)	0.62	1.17	0.97–1.32	0.91	0.73–1.06	below both
H3	skewness of $\Delta\mathcal{J}$	12.26	14.55	12.0–16.9	13.80	10.6–18.8	fails both
H3	up-jump var. share	0.838	0.904	0.87–0.92	0.828	0.77–0.87	fails both
H4	half-life (days)	120.1	91.7	82–111	161.6	122–200	*opposite signs*
H4	$\mathrm{sd}(\mathcal{J})$	2.03	1.61	1.38–1.92	2.17	1.45–2.78	i.i.d. only
H4	$\mathrm{range}(\mathcal{J})$	18.95	10.49	8.4–12.3	14.23	10.0–17.9	**clears both**
H5	ΔS_{mkt} gap	−0.053	−0.025	−0.100–0.040	−0.001	−0.051–0.081	edge (block)
H5	$\mathrm{corr}(\Delta\mathcal{J}, \Delta S_{mkt})$	−0.243	−0.067	−0.247–0.160	−0.013	−0.239–0.293	edge (both)
H6	coupling, stress	0.66	0.90	0.86–0.93	0.76	0.63–0.83	fails both
H6	coupling gap	0.617	0.746	0.64–0.88	0.667	0.54–0.75	fails both
–	$\mathrm{corr}(\mathcal{J}, \log \mathrm{vol})$	0.461	0.414	0.36–0.46	0.265	0.18–0.33	—

Table 2: Observed statistics against two calibrated nulls, twenty replications each. "Clears both" means the observed value lies outside all forty replications on the hypothesised side. Gaps are stress minus calm. Note that several statistics are *larger* in a null than in the data.

What clears both nulls. Two things, and only two. The regime difference in the *dependence* channel: −1.72 nats against −0.29 and −0.64, outside every one of forty replications. And the *range* of the structural index: 18.95 nats against 10.49 and 14.23. In words: what separates a crisis from a run of large returns is that the cross-section couples, and the structural index travels further in real markets than in any resampling of the same returns.

What does not. The jump asymmetry fails the i.i.d. null outright (H3). The channel coupling fails both, and fails in the direction that matters: the nulls couple *more* strongly than the data (H6). The shape channel's regime signal is smaller than both nulls. The compensation hypothesis is lost in the nulls' own noise (H2).

What is not identified. The relaxation half-life is 120 days against 92 under the i.i.d. null and 162 under the block null — the data sit *between* the two, and the sign of the comparison flips. Block resampling occasionally concatenates high-volatility blocks and manufactures long excursions; i.i.d. resampling manufactures short ones. The honest conclusion is that the level of the half-life is an artefact of whichever null one chooses to believe, and that P1 is supported by the amplitude result rather than by the half-life. The skewness signature (H5) sits at the fifth percentile of both nulls with the right sign, which is suggestive and not more.

One thing to note about $\mathcal{J}$ and volatility. Under the block null the correlation between $\mathcal{J}$ and log volatility is 0.27; in the data it is 0.46. The structural index tracks volatility *more* closely than a clustering-only process would, not less. Table 6 shows the shape channel is nevertheless orthogonal to volatility, so the extra tracking comes through the dependence channel — correlation and volatility genuinely move together in a way clustering alone does not reproduce.

And one thing none of this touches. The stress-testing construction of Section 8 is a measurement convention plus a convex optimisation. It does not rest on any of H1–H7 being true.

7.2 H1: the regime signature

Table 3 compares the top decile of trailing 21-day market volatility (807 dates) against the remainder (7,254). The block-bootstrap p-values are one-line summaries of a null in which the regime label carries no information about the level.

Channel	Stress	Calm	Difference	p
H^{vol} (scale)	−75.51	−80.98	+5.47	< 0.001
H^{dep} (dependence)	−5.12	−3.40	−1.72	0.001
H^{cov} (Gaussian ceiling)	−80.63	−84.37	+3.75	0.013
$\mathcal{D}$ (shape)	1.96	1.48	+0.48	0.033
asymmetry channel	0.121	0.112	+0.009	0.449
tail channel	1.837	1.364	+0.473	0.036
$\mathcal{J}$ (structural index)	8.02	5.13	+2.89	< 0.001
mean pairwise correlation	0.357	0.251	+0.106	0.001
effective number of modes	10.50	13.20	−2.70	0.001
tail co-exceedance rate	0.305	0.219	+0.086	0.002
market skewness	−0.188	−0.221	+0.033	0.671
market excess kurtosis	2.39	1.67	+0.72	0.141

Table 3: H1: channel means by regime, S&P 500 individual stocks, 1990–2022. Entropies in nats.

Every sign is as H1 predicts. Dependence entropy falls by 1.72 nats, the effective number of independent risk modes drops from 13.2 to 10.5 out of 20, the shape deficit rises by 0.48 nats, and the structural index rises by 2.89 nats – a 56% increase, the most significant difference in the table. The replication panel agrees on every sign, with $\mathcal{J}$ rising from 0.89 to 1.23 nats ($p < 0.001$).

Against the calibrated null the hypothesis splits in two. The *dependence* channel carries a real signal: the observed regime difference of -1.72 nats is roughly six times the null's -0.29 and lies outside all twenty replications. So does the difference in $\mathcal{J}$ itself, though by a much narrower margin (2.89 against a null mean of 2.32). The *shape* channel does not: its observed regime difference of $+0.48$ nats is smaller than the null's $+1.35$. Scrambled data, in which a "stress" window is by construction a window containing isolated extreme returns, fatten the measured marginals more than real stress episodes do. The rise in $\mathcal{D}$ under stress is therefore not evidence of anything beyond the presence of large returns.

The economically meaningful part of H1 is thus narrower than the raw table suggests, and it is the part one would want it to be: what distinguishes a real crisis from a run of large returns is that *the cross-section couples* — the effective number of independent risk drivers collapses — not that the marginals fatten.

Two entries in the table are as informative as the confirmations. The *level* of market skewness barely moves between regimes ($p = 0.67$): stress is not a state in which returns are persistently more left-skewed, a point we return to under H5, where the skewness signature appears at the moment of the jump rather than throughout the regime. And the Gaussian ceiling H^{cov} rises in stress, by 3.75 nats – because the scale channel dominates it. Differential entropy computed from a covariance matrix goes *up* in a crisis. Any attempt to use the level of Gaussian entropy as a stress indicator therefore has the sign backwards; this is precisely why the paper works with the scale-free $\mathcal{J}$.

7.3 H2: compensation does not survive the move to individual firms

An earlier draft of this project, on the 49 Fama–French industry portfolios, found $\mathrm{corr}(\Delta H^{vol}, \Delta H^{dep}) = -0.63$ over 1990–2026 and a 57% reduction in the variance of ΔH^{cov} relative to an uncorrelated-components benchmark: marginal volatility and dependence offset, leaving the covariance determinant markedly more stable than either component.

On individual stocks that result does not replicate.

	All dates	Calm	Stress
S&P 500, 20 individual stocks	$+0.10$	$+0.28$	-0.40
`EuStockMarkets`, 4 indices	-0.23	-0.13	-0.67
FF49 industry portfolios (earlier draft)	-0.63	-0.74	-0.40

Table 4: H2: $\mathrm{corr}(\Delta H^{vol}, \Delta H^{dep})$ by regime and dataset. Compensation is a stress phenomenon on individual firms and on a small index panel; the strong unconditional compensation reported for industry portfolios is not reproduced on firms.

Table 4 is a negative result for H2 as originally stated, and a useful one. Over the full sample the correlation on individual stocks is $+0.10$; in calm markets it is $+0.28$, the wrong sign entirely; only in stress does it turn to -0.40 and the variance of ΔH^{cov} fall 31% below its benchmark. Two readings are available and the data here cannot separate them. Industry aggregation may manufacture compensation: an industry portfolio is already a diversified bundle, so a common shock raises its volatility and its correlation with other bundles in a mechanically linked way that individual firms do not share. Or cross-sectional size may drive it, since 48 industries offer 1,128 pairwise correlations into which risk can be reallocated against 190 for twenty stocks. What can be said is that the unconditional compensation regularity should not be quoted as a general property of equity markets, and that an entropy diagnostic resting on it would not have transferred from industries to firms.

The consequence for the rest of the paper is the one that motivated Section 3: if the Gaussian ceiling is not reliably conserved and in any case rises with volatility, the informational content of a crisis has to be looked for somewhere other than $\det \Sigma$.

7.4 H3: the jump component is one-sided

Taken at face value the evidence looks overwhelming. On the primary panel the four-sigma detector finds 158 up-jumps against 9 down-jumps; up-jumps carry 83.8% of the variance of $\Delta\mathcal{J}$ against 0.3% for down-jumps; the skewness of $\Delta\mathcal{J}$ is +12.3. The replication panel agrees (27 against 5, skewness +6.7). Figure 1 is visibly sawtoothed, with sharp rises at 1997–98, 2002, September 2008, 2011, 2015, 2018 and a spike to 21 nats in March 2020, each followed by a slow decay — exactly the shape (9) predicts.

The asymmetry does not survive the i.i.d. null. Under it the detector finds *on average 148 up-jumps and 3 down-jumps*, with a skewness of $\Delta\mathcal{J}$ of +14.6 — larger than the observed +12.3 — and an up-jump variance share of 90%, again larger than observed. Every one-sidedness statistic in the data is at or inside that null.

The block null is kinder on one count and not on the others. Preserving volatility clustering lowers the null up-jump count to 129.5, so the observed 158 does lie beyond it; but the skewness of $\Delta\mathcal{J}$ (13.8) and the up-jump variance share (0.83) still bracket the observed values, and the mean up-jump is still larger in the null than in the data. A count that clears one null while every measure of the *shape* of the asymmetry fails both is not a result. H3 is **not supported**.

The explanation is not subtle once stated. $\mathcal{J}$ is a non-negative convex functional of estimated second, third and fourth moments. A single fat-tailed observation entering an exponentially weighted estimator pushes it up sharply, and the estimator's own memory then lets it decay smoothly. Sawtooth dynamics are what a fat-tailed marginal distribution produces when passed through this measurement device, whether or not information arrives in lumps. The visual appeal of Figure 1 is, to this extent, an artefact.

Two things are worth salvaging. First, the null is a strong one and arguably too strong for the postulate as stated: because i.i.d. resampling preserves the fat tails, it preserves the *lumps* of information and scrambles only their timing. What the test rejects is therefore not "information arrives in discrete parcels" but the weaker and more useful claim that *the observed one-sidedness of $\Delta\mathcal{J}$ carries information about arrival dynamics beyond what the marginal distribution already implies*. It does not.

Second, one comparison in Table 2 runs the other way and is informative. The largest observed up-jump is 3.21 nats against a null mean of 7.19, and the mean up-jump is 0.62 against 1.17: real entropy jumps are about *half* the size the scrambled world produces. The reason is that real extreme days arrive when the market is already agitated, so the local scale against which they are measured is already elevated; in the scrambled world they arrive out of a calm baseline and look correspondingly more surprising. That is volatility clustering showing up in the entropy measure — evidence for temporal structure, obtained from a statistic constructed to test something else.

7.5 H4: relaxation, against the estimator's own memory

Estimating (9) on non-jump dates only – so the diffusive channel is identified without contamination – gives $\kappa = 0.0058$ per day ($t = -3.75$, HAC, $p = 0.0002$), a half-life of **120 trading days** and a long-run level of 3.34 nats. The replication panel gives $\kappa = 0.0050$, a half-life of 139 days, $p = 0.023$.

The half-life is where the two nulls disagree most sharply, and the disagreement is instructive. Under i.i.d. resampling the same estimator returns 91.7 days on average, so the observed 120.1

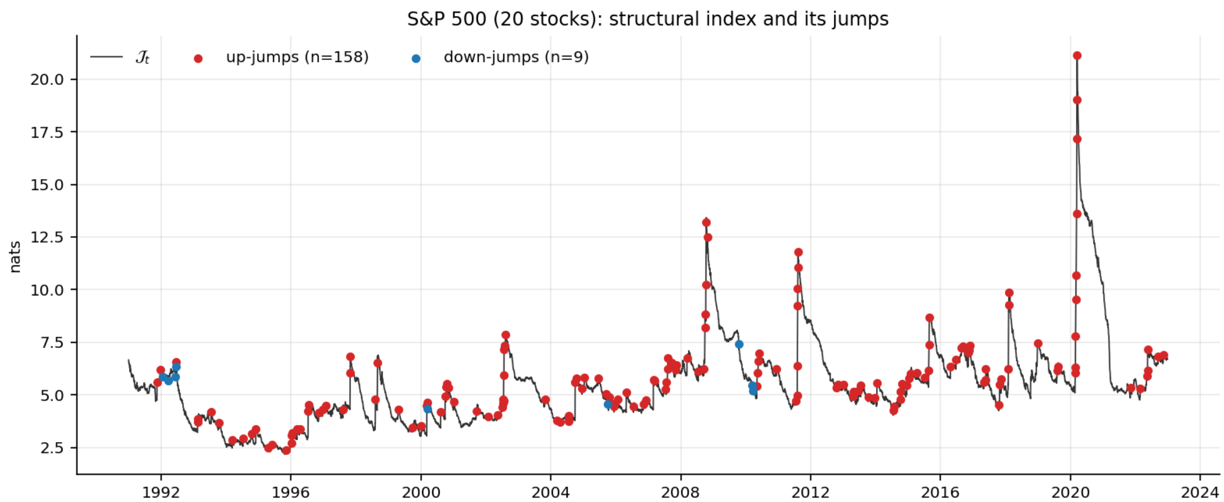

Figure 1: The structural index $\mathcal{J}_t$ on twenty S&P 500 stocks, 1990–2022, with detected jumps. Red points are entropy-destroying up-jumps, blue points down-jumps. The sawtooth — sharp rise, slow decay — is the qualitative content of (9).

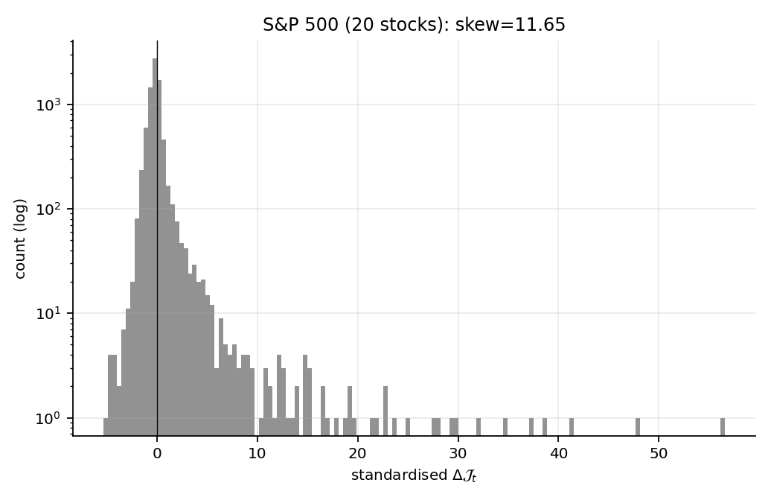

Figure 2: Distribution of standardised $\Delta\mathcal{J}_t$ (log counts). The right tail extends far beyond the left.

exceeds every replication. Under block resampling it returns 161.6 days, so the observed value falls *below* every replication. The data sit between the two nulls, and the sign of the comparison depends entirely on which one is asked. Block resampling occasionally concatenates high-volatility blocks and manufactures long excursions; i.i.d. resampling breaks up the runs that real markets contain and manufactures short ones. The level of the half-life is therefore **not identified**, and the 120-day figure should not be read as an absorption time — the controlled Gaussian simulation of Section 6.3, where a single shock decays in 35 days with no economics at all, sets the floor on how much of it is estimator memory.

What does survive is the *amplitude*. The range of $\mathcal{J}$ over the sample is 18.95 nats against 10.49 under the i.i.d. null and 14.23 under the block null, outside all forty replications. Its standard deviation clears the i.i.d. null (2.03 against 1.61) though not the block null. Real markets move the structural index further than any resampling of the same returns does, which is the part of P1 the data support: not a specific relaxation rate, but a state variable that genuinely travels.

Figure 3 adds a caveat the point estimate hides. Averaging the index around detected up-jumps shows that $\mathcal{J}$ begins rising about twenty days *before* the flagged date, peaks about thirteen days after it, and returns to the pre-event level after roughly eighty days. Information events cluster: a flagged jump is typically not an isolated arrival but one of a sequence. The clean single-jump-then-decay picture of (9) is therefore an idealisation, and a Hawkes-type self-exciting arrival process would fit the observed clustering better than the Poisson arrivals assumed here.

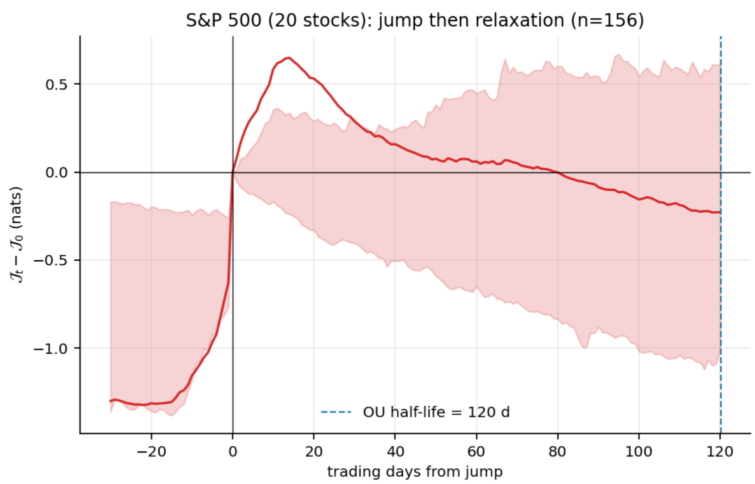

Figure 3: Average path of $\mathcal{J}_t$ around detected up-jumps, demeaned at the event date, with the interquartile band. The rise before day zero is jump clustering.

7.6 H5: the jumps carry a sign

Postulate P2 predicts that up-jumps arrive with negative market skewness. On up-jump dates the change in market skewness averages −0.052 against +0.001 elsewhere; the level of market skewness on those dates is −0.394 against −0.214; and $\text{corr}(\Delta\mathcal{J}, \Delta S_{mkt}) = -0.24$. The asymmetry channel of the deficit rises on jump dates (+0.010 against −0.0002), and so does tail co-exceedance (+0.008 against −0.0002, correlation with $\Delta\mathcal{J}$ of +0.35). The replication panel is stronger on this dimension: $\text{corr}(\Delta\mathcal{J}, \Delta S_{mkt}) = -0.38$ and a mean skewness change of −0.129 on up-jump dates.

The qualification already visible in Table 3 is that the skewness signature is an *event* property, not a regime property: sorting dates by volatility finds no significant skewness difference, sorting by entropy jump does.

Against the nulls, H5 is suggestive and no more. Both produce the effect in the same direction without any economics, because the mechanical link is obvious once stated: a large negative return simultaneously raises $\mathcal{J}$ and lowers the measured skewness of the window it enters. The observed correlation of -0.243 sits at roughly the fifth percentile of both nulls (-0.067 i.i.d., -0.013 block), and the mean skewness change on jump dates does the same under the block null while sitting well inside the i.i.d. one. We report H5 as *not established*: the direction is right, it is consistent across two nulls and two datasets (the replication panel gives -0.38), but a p-value of 0.05 to 0.10 on twenty replications is a reason to look further, not a finding. Postulate P2 remains a postulate.

7.7 H6: the channels lock together under stress

corr $\left(-\Delta H^{dep}, \cdot\right)$	All	Calm	Stress
S&P 500: with $\Delta\mathcal{D}$ (shape)	0.34	0.04	0.66
S&P 500: with the tail channel	0.34	0.03	0.68
S&P 500: with the asymmetry channel	0.06	0.15	-0.01
`EuStock`: with $\Delta\mathcal{D}$	0.23	0.05	0.54

Table 5: H6: coupling between the two entropy-destroying channels, by regime.

In the data the pattern is striking. In calm markets the dependence channel and the tail channel are effectively unrelated (0.03); under stress their correlation is 0.68. The replication panel reproduces it at 0.05 against 0.54, and the coupling is carried entirely by the *tail* channel, with the asymmetry channel not participating.

Both nulls reproduce it, and both reproduce it *more strongly* than the data: stress coupling of 0.90 (i.i.d.) and 0.76 (block) against the observed 0.66, and a stress-minus-calm gap of 0.75 and 0.67 against the observed 0.62. The mechanism is mechanical rather than economic. A window containing a large common return has both a higher measured correlation and fatter measured marginals, so the two channels move together in any data in which large days exist, scrambled or not. H6 is **not supported** as evidence for a common binding constraint.

Adding the block null does change the interpretation of the gap between null and data. Under i.i.d. resampling the null coupling is near-deterministic (0.90, s.d. 0.03); preserving clustering brings it down to 0.76 with s.d. 0.07, closer to but still above the observed 0.66. Real stress episodes couple the two channels *less* tightly than either null. If that is not noise it points the opposite way from Section 4: dependence and tail weight would have partly independent drivers in real crises, which a common-constraint story does not predict.

Over the full sample the variance of $\Delta\mathcal{J}$ splits 40% to dependence and 60% to shape on individual stocks, and 58%/42% the other way on the four-index panel – a reminder that the balance between the channels depends on the cross-section being measured.

7.8 H7: no incremental forecasting value

H7 is rejected. Under expanding-window walk-forward estimation with annual refitting, evaluation on non-overlapping dates and Diebold–Mariano tests, no entropy channel – $\mathcal{J}$, the deficit $\mathcal{D}$, the dependence term, the crisis indicator or the unhealed-share indicator – significantly improves out-of-sample forecasts of 21-day-ahead realised volatility, drawdown or worst daily loss relative to a model using the scale channel alone. The largest improvement anywhere is 3.9% in mean squared error, with $p = 0.52$; several specifications are significantly *worse*. In-sample coefficients are highly significant ($p < 10^{-4}$ for $\mathcal{J}$ in every specification), which is exactly the pattern one expects from overlapping windows and is why the in-sample numbers are not reported as evidence.

This replicates, on a different dataset and a much more demanding evaluation protocol, the negative predictive result the earlier draft of this project obtained. Two conclusions follow. The honest one is that the entropy channels are diagnostic rather than predictive: they describe the informational state of the market well and do not forecast its next move under a linear specification. The constructive one is that the value of the framework has to be sought where Section 8 looks for it – in the internal consistency and the calibration of scenarios, not in point forecasts.

7.9 Is any of this just volatility?

Correlation with log trailing volatility	S&P 500	`EuStock`
H^{vol} (scale channel)	+0.56	+0.45
$-H^{dep}$ (dependence channel)	+0.56	+0.66
$\mathcal{D}$ (shape channel)	+0.09	−0.06
$\mathcal{J}$ (structural index)	+0.46	+0.60
tail co-exceedance	+0.32	+0.49

Table 6: How much each channel shares with volatility.

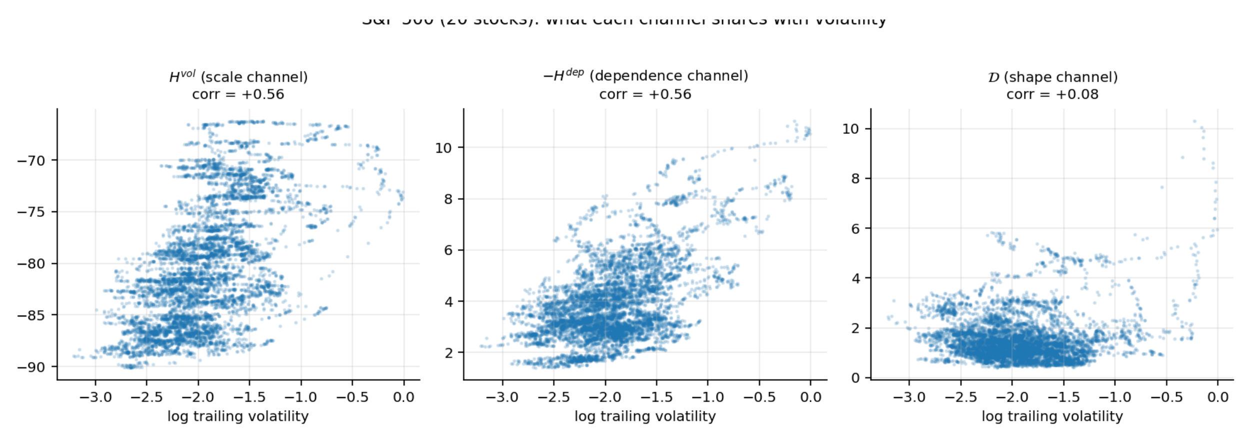

Figure 4: Each channel against log trailing volatility. The scale and dependence channels track it; the shape channel does not.

Table 6 and Figure 4 answer the obvious objection. The structural index shares about a fifth of its variance with volatility on individual stocks ($r = 0.46$), so it is related but far from redundant. The shape channel is *orthogonal* to volatility ($r = 0.09$ and -0.06): the fat-tailedness of the conditional distribution is a dimension of market state that a volatility measure does not see at all. Since the shape channel carries 60% of the variance of $\Delta\mathcal{J}$ on the primary panel, most of what the index moves on is information volatility does not contain.

7.10 Replication

Figure 5 runs the identical pipeline on the four European indices, 1991–1998. The panel is far too small to carry a result on its own — four assets give six pairwise correlations against 190 for the primary panel, and one stress cycle — but every sign reported above reproduces: $\mathcal{J}$ rises in stress

(1.23 vs 0.89, $p < 0.001$), the channels couple under stress and not in calm (0.54 against 0.05), the index relaxes with a half-life of 139 days, and up-jumps carry a mean market-skewness change of -0.129 against $+0.003$ elsewhere.

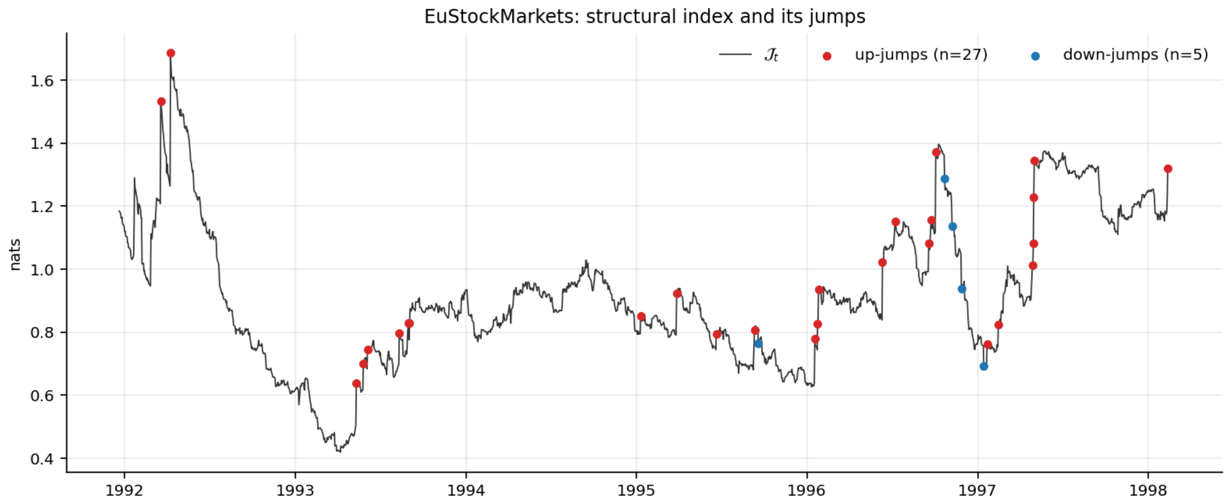

Figure 5: Replication: the structural index on the DAX, SMI, CAC and FTSE, 1991–1998. The visible rises coincide with the September 1992 and 1993 ERM episodes and the October–November 1997 Asian crisis.

8 Stress testing by entropy

8.1 The construction

Definition 2 (Entropy-ball scenario)**.** Given today's conditional distribution p_0 and a severity $\eta > 0$, the scenario of severity η for a risk functional ρ is

$$p^\star_\eta \;=\; \arg \max_{p\,:\,D_{KL}(p\,\|\,p_0)\,\le\,\eta} \rho(p). \tag{10}$$

For a linear portfolio and a Gaussian reference only the one-dimensional projection matters. With m_0, s_0 the portfolio mean and standard deviation and $u = (m - m_0)/s_0$, $v = s/s_0$, the constraint is $g(u,v) = \frac{1}{2}\big(u^2 + v^2 - 1 - 2\log v\big) \le \eta$, and for any ρ of the form $-m + cs$ – Gaussian VaR and expected shortfall are both of that form – the solution satisfies $u = -t$, $v - v^{-1} = ct$, with t fixed by $g = \eta$. The special case $c = 0$ is the sanity check worth remembering:

$$\text{a pure directional scenario of } \eta \text{ nats is exactly a } \sqrt{2\eta}\text{-sigma move.} \tag{11}$$

One nat buys 1.41 sigma. This is the exchange rate between information and risk, and it is the whole reason the construction is usable by a committee: the severity parameter is not an abstraction.

8.2 Calibrating the radius

The radius is not a modelling choice here. Define the h-step information flow

$$I^{(h)}_t \;=\; D_{KL}(p_t\,\|\,p_{t-h})\,, \tag{12}$$

the divergence between the market's conditional distribution today and h days ago – how much the market revised its own view. This is a proper divergence between two conditional return distributions, measured in the same unit as the scenario radius, which is what makes the calibration coherent. Quantiles of $I^{(21)}$, thinned to non-overlapping blocks, convert a return period into a radius.

Return period	1y	2y	5y	10y	20y	50y†
radius η (nats)	1.04	1.63	2.34	3.88	7.54	13.16
directional shock (σ)	-0.58	-0.72	-0.84	-1.06	-1.45	-1.88
volatility multiplier	2.04	2.33	2.63	3.15	4.10	5.20
annualised portfolio vol	41%	47%	53%	63%	82%	105%
one-day ES_{99}	7.6%	8.8%	9.9%	12.0%	15.6%	19.9%

Table 7: Severity ladder for an equally weighted portfolio of the twenty stocks, calibrated on 383 non-overlapping 21-day information-flow observations. Today's baseline is 20% annualised volatility and a one-day ES_{99} of 3.4%. † beyond the sample: the 50-year rung requires a quantile the 33-year history cannot resolve and is an extrapolation of the empirical tail.

Two features of Table 7 are worth reading carefully. The scenario spends its budget mostly on *variance* rather than on direction – at the one-year rung, a doubling of volatility and only a 0.58σ drift shock – because expected shortfall is more sensitive to scale than to location. And the radius is not comparable across universes: a KL divergence grows with the dimension of the return vector, so the four-index panel's ladder runs from 0.09 to 0.21 nats where the twenty-stock ladder runs from 1.04 to 13.2. A ladder must be calibrated for the portfolio it will be applied to.

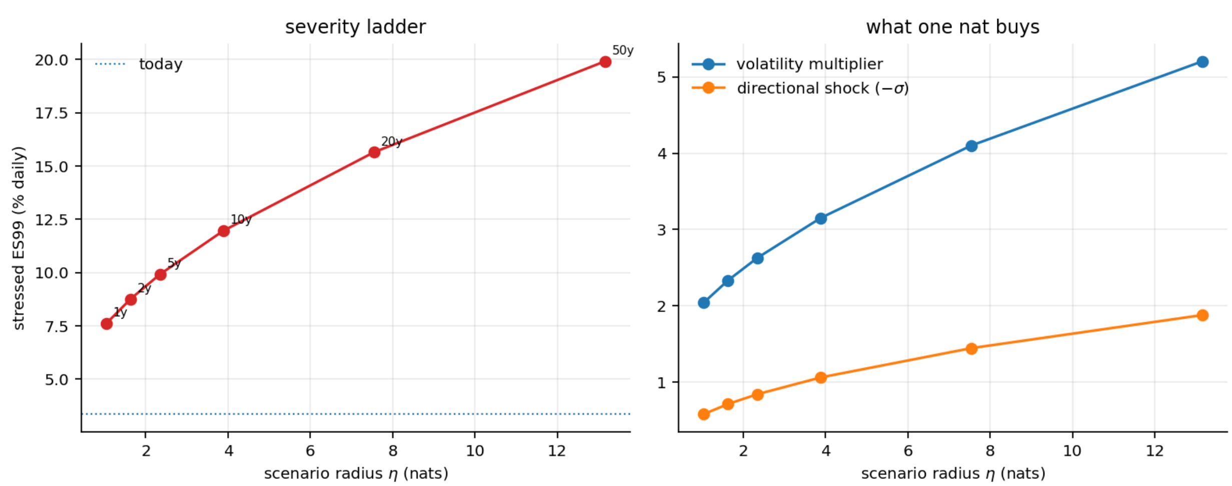

Figure 6: Left: stressed one-day ES_{99} against scenario radius, with return periods marked. Right: what one nat buys, split between the volatility multiplier and the directional shock.

8.3 What this adds over volatility, mean and skewness stress tests

The comparison is not rhetorical, because the same metric prices a classical scenario. Take today's conditional distribution and build the textbook bundle: marginal volatilities multiplied, all pairwise correlations forced to a single high number, a directional shock of several sigma. Its informational price is $D_{KL}(p_{\text{classical}} \,\|\, p_0)$, computable in closed form.

Classical bundle	mild	standard	severe
volatility multiplier	1.5	2.0	3.0
all correlations forced to	0.70	0.90	0.95
directional shock	-2σ	-3σ	-4σ
implied one-day ES_{99}	9.1%	13.7%	20.3%
informational price (nats)	6.80	9.97	20.19
of which: directional leg	2.00	4.50	8.00
volatility leg	4.39	16.14	58.03
correlation leg	2.43	8.10	13.50
implied return period	$\sim$ 11y	$>$ 30y	beyond sample
entropy scenario at the same $\mathbf{ES_{99}}$	**1.85**	**5.47**	**13.74**
its return period	2.3y	10.6y	$\sim$ 32y
informational overpayment	73%	45%	32%

Table 8: Pricing textbook stress bundles against the market's own information flow, and comparing each with the entropy scenario that delivers the same expected shortfall. Reference rungs from Table 7: the twenty-year scenario is 7.54 nats, the fifty-year scenario 13.16; the worst single day in 33 years prices at 9.03.

Table 8 is the applied point of the paper, and it says something sharper than "classical scenarios are too severe". The standard bundle costs 9.97 nats, which sits between the twenty-year and fifty-year rungs – a defensible severity for a regulatory stress test, and one nobody at the table had any way of knowing they had chosen. What is not defensible is the efficiency. An entropy scenario delivers the same 13.7% expected shortfall for 5.47 nats. The committee believes it has specified something on the order of a thirty-year event; measured by the loss it actually produces, it has specified a ten-year event and paid thirty-year implausibility for it. The mild bundle is worse: it prices like an eleven-year event and delivers a *two-year* loss, a 73% overpayment.

Where does the money go? The directional leg is cheap and exactly priced: shifting the portfolio mean by k standard deviations costs $k^2/2$ nats, so 4.5 for three sigma. The expensive leg is the uniform doubling of every marginal volatility, at 16.1 nats on its own – more than the entire bundle. Forcing all correlations to 0.9 costs 8.1 nats on its own.

The legs are not additive, and the way they fail to be is instructive: the three sum to 28.7 nats while the bundle costs 9.97. Forcing correlations upward collapses the covariance onto a near-rank-one subspace that the reference distribution already grants generous variance – the market mode – so scaling that collapsed structure up is far cheaper than scaling the full-rank one ($\mathrm{tr}(\Sigma_0^{-1}\Sigma_1)$ falls from 20 to 10.5 when the correlations are forced, before the volatility multiplier is applied at all). The volatility shock and the correlation shock partly pay for each other.

This is exactly the structure H6 identified in the data. Real crises do not equalise correlations and inflate all volatilities independently; they load the cross-section onto one or two collective modes and fatten the tails of the marginals together. A framework that shocks volatility, correlation and shape as three separate scalars has no way to represent that, and no way to notice that its bundle is buying severity at a 30–70% markup.

Four concrete advantages over the volatility/mean/skewness approach follow.

1. **One severity scale, in a unit that does not depend on the asset.** Scenarios across desks, portfolios and asset classes become comparable, and severity is quoted with a return period rather than an adjective.

2. **Internal consistency by construction.** The location and scale shocks come from one budget and one optimisation. A bundle that shocks volatility, correlation and skewness independently can, and often does, describe a distribution the market could not produce.

3. **Existing scenarios can be audited rather than replaced.** Any regulatory template or committee judgement gets a price in nats and a return period, so the question of whether a stress test is calibrated becomes arithmetic. This is the practical payoff of Table 8.

4. **Reverse stress testing comes free.** Instead of asking what a scenario does, ask what the cheapest scenario producing a given loss is. On the primary panel, the empirical 99% one-day loss (3.2%) costs 1.5 nats; the 99.9% loss (7.7%) costs 4.8; the worst day in 33 years (11.5%) costs 9.0. Reading those against Table 7 converts a loss the board cares about into a frequency.

A caution on the fourth item. Reverse stress tests computed non-parametrically, by tilting the empirical sample, degenerate as the radius grows: at 9.0 nats the tilted measure has an effective sample size of 1.0, meaning it is the single worst day wearing a distribution's clothes. The implementation reports the effective sample size for exactly this reason, and the Gaussian-ball construction should be used whenever it falls below a few dozen.

8.4 Two indicators for monitoring

The jump/diffusion split yields two quantities that can be produced daily at negligible cost. The *crisis* indicator is the standardised positive-jump component of $\mathcal{J}$: how much information arrived today, in units of the local diffusive scale, zero when no jump is detected. The *unhealed* indicator is $\mathcal{J}_t$ as a share of its trailing maximum: how much of what already arrived has not yet relaxed away. Under P1 it should decay with half-life $\ln 2/\kappa$; a market where it stays near one for materially longer is one where the shock has not been absorbed. Neither improves a volatility forecast (H7); both describe a state that volatility does not.

8.5 Relevance to insurance solvency

Three points connect this to the insurance side of the wider project. First, an entropy-ball scenario is a natural ORSA object: it produces a full stressed distribution, not a shock vector, so it feeds a capital calculation directly and carries a defensible plausibility statement. Second, the leg decomposition of Table 8 bears on the Solvency II Standard Formula, whose diversification credit is built from a fixed correlation matrix: shocking volatilities and correlations as separate scalars, which is the usual conservative adjustment, buys a given tail loss at a 30–70% markup in plausibility terms, because it does not represent the way observed crises redistribute dependence – onto a few collective modes rather than uniformly. Third, the coherence properties of entropic risk measures (Ahmadi-Javid, 2012; Artzner et al., 1999) make the $\mathcal{J}$ framework compatible in principle with internal-model capital requirements, though nothing here has been tested on liability portfolios and the short, low-frequency panels typical of insurance data would strain every estimator used above.

9 Limitations

The bridge is an argument, not a proof. Section 2.5 lists the failures of linear impact and sign exchangeability. Nothing empirical below depends on the volume clock, but the framing does.

The deficit is estimated marginally. The shape channel captures marginal non-Gaussianity and Gaussian-copula dependence; non-linear tail dependence is measured separately rather than integrated, so $\mathcal{J}$ understates the true divergence, by an amount that is itself largest under stress.

The mechanical null swallows one hypothesis whole. The jump asymmetry is entirely reproduced by i.i.d. data with the same marginal distribution, and about two-thirds of the measured relaxation is estimator memory. The persistence and amplitude results survive; the asymmetry result does not, and no reader should quote the 158-against-9 figure without the null beside it. Twenty replications is also few: the null quantiles in Table 2 are themselves estimated with visible error, and a hundred replications would sharpen the p-values that currently sit at 0.05.

Poisson arrivals are the wrong model. Figure 3 shows clustering that (9) does not accommodate. A Hawkes self-exciting arrival process is the obvious next specification.

Twenty stocks is a small cross-section. The dependence channel is sensitive to N, the balance between channels shifts with it, and the KL radius is not comparable across universes. Results should be reproduced on a wide cross-section — the Kenneth French industry file, or CRSP — before any number here is treated as a constant of nature.

The severity ladder extrapolates in the tail. With 383 non-overlapping information-flow observations, rungs beyond about twenty years are interpolations of the empirical tail, not measurements. Fitting a generalised Pareto tail to $I^{(h)}$ would be the disciplined fix.

Nothing here forecasts. H7 is rejected on two datasets under a demanding protocol. Any use of these measures should be diagnostic.

10 Conclusion

The theoretical contribution is a chain that is explicit about which link is a theorem and which is an assumption. Maximum entropy under first and second moments gives a Gaussian; applied in volume time, with traded volume supplying the variance budget, it gives Gaussian returns conditional on volume, and temporal consistency turns those into a subordinated Brownian motion and a geometric Brownian price. Read backwards, non-Gaussian prices measure the distance from maximal ignorance, and that distance decomposes into a dependence channel and a shape channel whose sum, the structural index $\mathcal{J}$, is scale-free and non-negative. Conditioning cannot raise expected entropy, so entropy falls in jumps whose size is the mutual information of the news; that entropy recovers diffusively, and that the fall is one-sided in the loss direction, are the two postulates the paper adds and tests.

The empirical picture is mostly a warning, and we think that is the useful result. Every descriptive claim the framework makes is visible in the data: stress raises the structural index, the index is sawtoothed, jumps are one-sided, they carry negative skewness, and the dependence and tail channels couple under stress. Passed through a null that keeps the marginal distribution and scrambles the timing, most of it disappears — and several statistics come out *larger* in the scrambled data than in the real data. A non-negative convex functional of estimated second, third and fourth moments rises sharply and decays smoothly whenever a fat-tailed observation enters the estimator, and it does so whether or not information arrives in parcels. That caution applies to any entropy-based crisis indicator, not only to this one, and we have not found it stated in the applied literature.

Exactly two statistics clear both nulls, and they are the two worth keeping. What separates a crisis from a run of large returns is that the cross-section couples: the dependence channel's regime signal is between three and six times the nulls', while the shape channel's is smaller than both. And the structural index travels further over the sample than in any resampling of the same returns. The relaxation rate itself is not identified — it sits between the two nulls — so P1 is supported by amplitude rather than by a half-life. Two further results go against the paper's interest: the volatility–dependence compensation obtained on industry portfolios does not survive the move to individual firms, and no entropy channel forecasts anything out of sample.

The applied contribution is what the framework buys once one stops asking it to forecast. Measuring scenario severity in nats gives stress testing a unit, calibrating the unit on the market's realised information flow gives it a frequency, and solving for the worst distribution within the resulting ball gives a scenario that cannot be internally inconsistent. Applied in reverse it prices the scenarios institutions already run: on this sample, a standard textbook bundle turns out to sit four times beyond the fifty-year rung, and its cost is dominated by the one leg – forcing all correlations to a single number – that observed crises do not actually take.

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